

A facile approach to direct growth of layer-tunable graphene on Ge substrates

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ABSTRACT

Direct synthesis of graphene with tunable layer number on semiconductor has proven challenging. Synthesis of layer-tunable graphene on catalytic metal surfaces, in particular Cu and Ni, has shown great success. The growth methods on metal layers involve an inevitable transfer procedure, which degrades and contaminates the graphene. Thus, direct synthesis of graphene on semiconductor is necessary and profound. Here, a facile synthesis approach for direct growth of graphene on Ge(110) substrate via ion implantation was reported. Interestingly, the graphene growth during annealing is not self-limiting and the thickness of graphene can be precisely controlled by the carbon ion implanted fluence. Moreover, a detailed growth process upon segregation was investigated, and the results show graphene nucleation tends to occur near the Ge atomic step edges. Furthermore, the facile approach involving the ion implantation can be adopted to investigate the growth process on semiconductor and to control the number of graphene layers, thus promoting the practical application of graphene in nanoelectronics.

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1. Introduction

Graphene is a two-dimensional material made up of carbon atoms densely arranged in a hexagonal lattice [1]. It has attracted interest from many researchers due to its superior properties in physics, chemistry and material science [2–6]. In particular, the graphene-based transistors in semiconductor industry have developed rapidly and now are considered as an option for post-silicon electronics [7]. There are two different approaches to integrate the graphene into transistors in fabricating devices: the first one is to transfer the graphene from the metal growth substrate onto a device-compatible substrate, and the second one is to directly grow graphene on a desired substrate [8]. However, the transfer process that usually introduces some defects (*i.e.* cracks, holes, or wrinkles) and contamination (organics or metal) will degrade the graphene-based device performance [9]. Thus, the direct synthesis of layer-tunable graphene on the device-compatible substrate is transfer-free and very attractive.

It is difficult to directly grow graphene on Si substrate due to low carbon diffusivity on Si surface [10] and relatively strong Si-C bonds [11,12]. Unlike Si, Ge does not form stable carbide at high temperature according to the phase diagrams of Ge-C system [13] and has ultralow carbon solid-solubility [14]. In addition, Ge is also considered as a promising electronic material due to its higher carrier mobility and compatibility with silicon-based semiconductor industry [15]. In recent years, many researchers attempt to grow graphene on Ge or Ge/Si substrates via various approaches [10,11,15–19]. The recent studies indicated that the synthesis of graphene on Ge substrate seems to be a self-limiting and surface-mediated process similar to Cu-catalyzed growth of graphene in the chemical vapor deposition (CVD) process [10,15,20]. However, the underlying mechanism that governs growth process of graphene on Ge substrate has not been well understood and reported.

Except for the CVD method, ion implantation is also a common way to introduce carbon atoms to the metal catalyst substrate, such as Cu [21,22] and Ni [23–25]. In recent years, ion implantation has been proposed to tailor the layer of graphene taking advantage of the low carbon solid-solubility in the metal catalyst and the precise controllability of implanted fluence of ion implantation technology. However, the carbon segregation is uncontrollable in

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polycrystalline catalyst due to dramatic changes of its grain in the annealing process, leading to the synthesis of defective and heterogeneous graphene [23,26]. Furthermore, the short-circuit diffusion of carbon through the grain boundaries to the catalyst surface can further degrade the graphene homogeneity [26,27]. On the basis of understanding of the segregation process, single crystal Ge was adopted to avoid all these shortcomings. Graphene grown on Ge (001) substrate was previously investigated, and it was found that the faceting of the underlying Ge would destroy the smoothness of surface in the graphene growth process [19,28–30]. According to the study of electronic and mechanical properties of graphene on different orientation of the germanium crystal by Kiraly et al., Ge(110) substrate was more preferable and remained flat for graphene growth [28]. The carrier mobility value of the graphene grown on Ge(110) was also higher than that of growth on Ge(001) [18]. Therefore, single crystal Ge(110) is an ideal substrate for exploring the possibility to grow graphene via ion implantation.

In this work synthesis of layer-tunable graphene on single crystal Ge(110) substrate via ion implantation has been demonstrated. The graphene growth process was investigated by a series of experiments. Different from synthesis graphene on Ni bulk where the carbon atoms segregate in the cooling stage, the segregation and self-assembly process of carbon in the Ge substrate occurred at the steady annealing period. In addition, the experimental results show that graphene nucleation tends to occur near the Ge atomic step edges and graphene growth on Ge substrate via ion implantation is not self-limiting.

2. Experimental section

2.1. Growth of graphene

Graphene was grown on Ge(110) substrate by carbon ion implantation. Firstly, the Ge(110) substrate (N-type Sb-doped, $\langle 110 \rangle \pm 0.1^\circ$, single side polished) was cleaned with acetone, alcohol and deionized water. Prior to the carbon ion implantation, the Ge substrate was dipped into 10% diluted HF to remove the native oxide. Then, carbon ion implantation was carried out at room temperature (RT) with the ion energy of 10 keV and the fluence of 1×10^{16} ions cm^{-2} on Ge(110) substrate by employing a home-built ion implanter (BNU-400) at Beijing Radiation Center. The implanted samples were placed in a commercially available tube furnace (Xiamen G-CVD Material Technology Co., Ltd.). The quartz tube was flushed with 500 sccm Ar and pumped below 1 Pa to remove air prior to the temperature being ramped. The furnace was then heated to the desired growth temperature (800–920 °C) in ambient pressure with a gas mix of Ar and H_2 (100:3 sccm gas flow rate). The growth temperature was close to the melting point of Ge and monitored via a thermocouple attached to the quartz tube. After annealing, the furnace was cooled down to room temperature by sliding the heated zone of the furnace away from the sample.

2.2. Graphene transfer

The as-grown graphene was transferred onto the target substrate by a standard polymethyl methacrylate-assisted (PMMA, Aldrich Prod. Nos. 445746, dissolved in anisole with a concentration of 40 mg/mL) wet transfer method using an etching solution ($\text{H}_2\text{O}/\text{HF}/\text{H}_2\text{O}_2$ with concentration 20:1:1) to dissolve the Ge substrates. After etching the Ge substrates, the free-standing PMMA/graphene film was washed and lifted out of the ultrapure water onto the target substrate. Finally, the PMMA layer was removed by hot acetone and blow dried with N_2 gas.

2.3. Characterization of graphene

The prepared samples were thoroughly characterized by optical microscopy (Olympus BX51), Raman spectroscopy (Horiba Jobin Yvon, with laser excitation at 532 nm, and power of 1 mW), SEM (FEI Nova NanoSEM430, 3 kV), TEM (Tecnai G2 F20 s-Twin), XPS (AXIS Supra/Ultra, using a monochromatic Al $K\alpha$ X-ray source), EBSD(AJEOLJSM-6500F, 20 kV), AFM (Agilent 5500, performing the topography measurement with silicon probes (PPP-NCH, Nanosensors) under tapping mode) and LFM (Asylum Cypher VRS, high-resolution atomic scale imaging with silicaon nitride probes (TR400PB, Asylum Research) under lateral mode in ambient conditions (20–25 °C, relative humidity 10–15%)).

3. Results and discussion

Fig. 1 is a schematic illustration of direct synthesis of graphene on Ge substrate by ion implantation. The carbon ions were implanted into Ge(110) substrates with the energy of 10 keV and the predetermined fluence. According to the Stopping and Range of Ion in Matter (SRIM) program [31], the average projected range of the implanted carbon ions is about 27 nm with the longitudinal straggling about 15 nm in Fig. S1. The fluence of implanted carbon ions can be precisely controlled to be 5×10^{15} ions cm^{-2} , 1×10^{16} ions cm^{-2} , 2×10^{16} ions cm^{-2} , which are corresponding to the carbon atomic density of single layer graphene, bilayer graphene and multilayer graphene, respectively. The carbon atoms implanted into Ge substrate act as carbon source and diffuse towards the surface of Ge substrate at an elevated temperature. In order to minimize the evaporation rate of germanium, the annealing treatment was conducted in atmospheric pressure, as described in Fig. 1 as well.

As a versatile tool, Raman spectra are used widely to study and characterize the properties of graphene. The Raman spectrum of

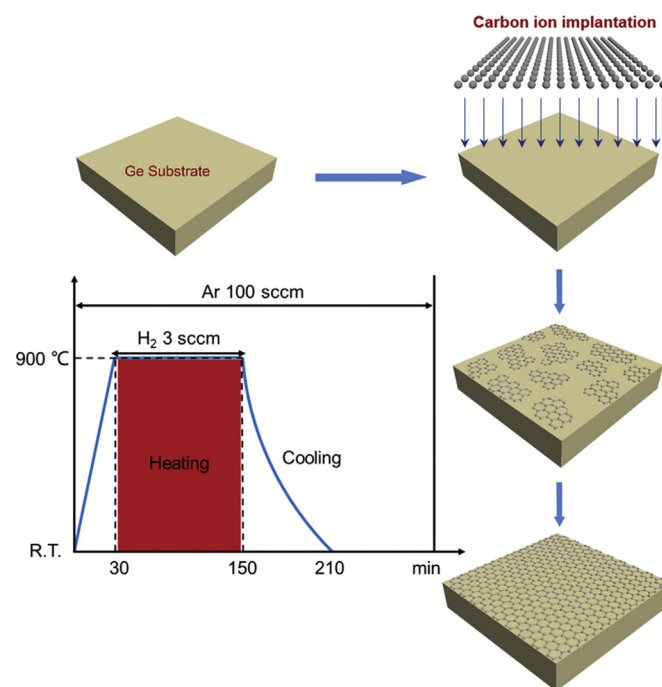


Fig. 1. Schematic illustration of direct growth of graphene on Ge(110) substrate via ion implantation. Carbon ions were implanted into Ge substrate followed by annealing in ambient pressure with a gas mix of Ar and H_2 . (A colour version of this figure can be viewed online.)

graphene usually consists of three characteristic peaks, D band at $\approx 1350\text{ cm}^{-1}$, G band at $\approx 1580\text{ cm}^{-1}$, and 2D band at $\approx 2680\text{ cm}^{-1}$. The relatively low D peak is known as the defect band or the disorder band. The G peak is the primary mode representing the planar configuration sp^2 bonded carbon. The 2D peak is more commonly referred to as an overtone of the D-band and usually used to determine graphene layer thickness [32]. In order to evaluate the effect of the annealing temperature and the growth time on the properties of graphene growth on Ge substrate, Raman spectra were performed on the samples under the annealing temperatures (800–920 °C) for 120 min, and the growth time for 30–180 min under the annealing temperature of 900 °C. As shown in Fig. 2a, no graphene formation was observed on the Ge surface for the temperature below 800 °C. It is worth noting that the D peak decreases and 2D peak rises gradually with the increase of the annealing temperature at the same growth time of 120 min, indicating the improved quality of graphene obtained at higher temperature for certain time. As the annealing temperature was close to the melting point (938 °C) of Ge, the high-quality graphene can be obtained on the Ge surface above 900 °C. Raman mapping images of graphene were conducted in a $30 \times 30\text{ }\mu\text{m}^2$ area to characterize the uniformity of graphene. In Fig. 2c, more than 97% of the area has I_D/I_G ratios of 0.3–0.4 and more than 95% of the area has I_{2D}/I_G ratios ranging from 0.8 to 1.2 in Fig. 2d, suggesting the formation of uniform bilayer graphene. Considering uniformity of graphene, the annealing temperature of 900 °C was more favorable and adopted in the direct synthesis of graphene on Ge substrate

because high evaporation rate may destroy the flatness of surface of Ge under higher temperature, as shown in Fig. S2. The results in Fig. 2b showed that thermal annealing experiment at 900 °C for 120 min was an appropriate condition in the growth of graphene on Ge substrate. It is noted that I_{2D}/I_G intensity ratio of 0.8–1.2 indicates the formation of bilayer graphene, which is corresponding to the implanted fluence of $1 \times 10^{16}\text{ ions cm}^{-2}$. Based on the above study, annealing temperature and growth time play crucial roles in determining the quality of graphene.

In order to investigate the growth process of graphene, the evolution of the graphene on Ge surface at different growth stages was intentionally characterized by SEM and AFM. Fig. 3 clearly display the different growth stages of the sample with implanted fluence of $1 \times 10^{16}\text{ ions cm}^{-2}$ for 30 min, 60 min and 120 min at 900 °C, respectively. As shown in Fig. 3a, the well-aligned graphene island (like a feather) arrays display the same orientation, which was indicated by the white dashed lines. Moreover, there is a preferred growth direction along $[-110]$ at the early stage of growth on Ge(110) substrate, whose direction coincides the armchair direction of the graphene lattice. As the graphene islands grew, the same orientations of adjacent graphene islands were merged together, which were pointed out by the white arrows in Fig. 3b. Finally, the graphene islands almost grew on the entire surface of the Ge substrate in Fig. 3c. It should be noting that most commercially available Ge substrates are not perfect and have some atomic steps and flat terraces resulting from the wafer polishing at an off-axis angle [33]. Furthermore, there was a very interesting

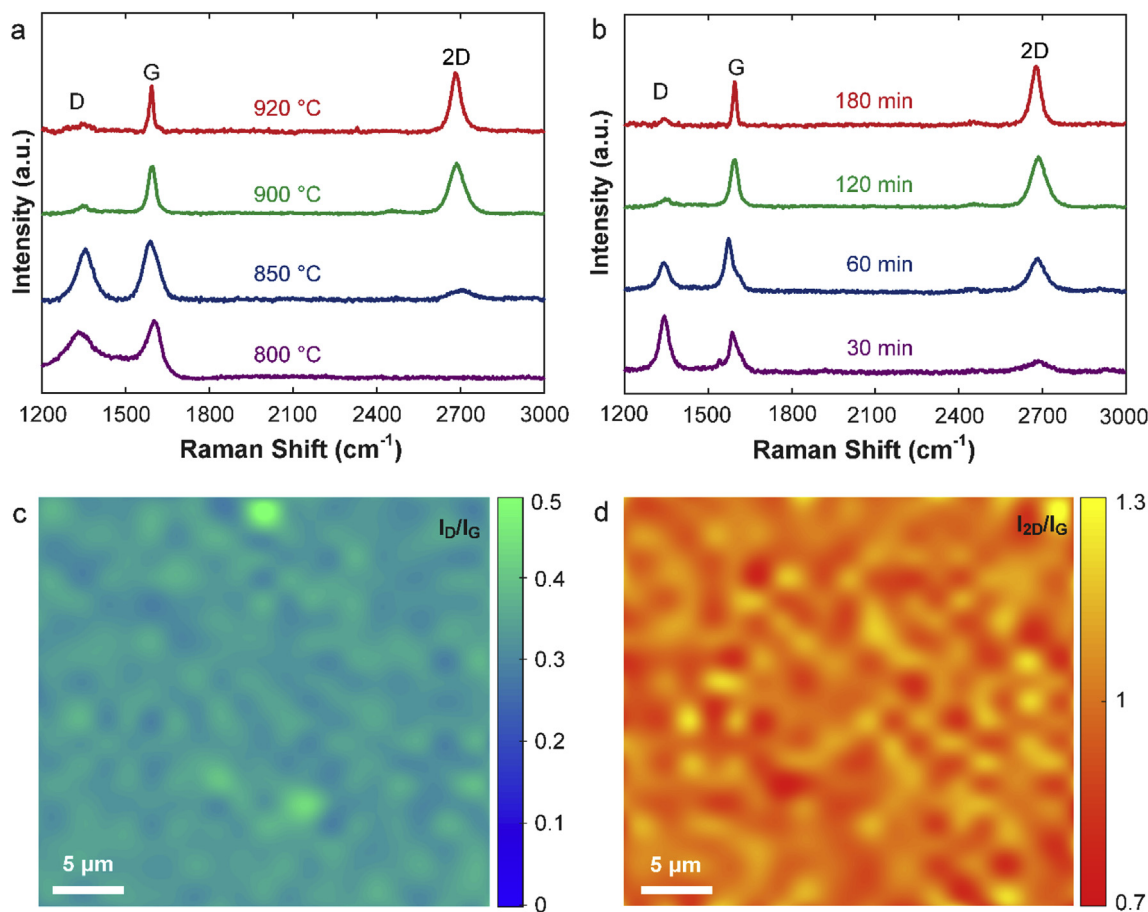


Fig. 2. (a) Raman spectra of the temperature-dependent growth of graphene on Ge substrate for 120 min; (b) Raman spectra of the time-dependent growth of graphene on Ge substrate at 900 °C; (c) (d) Raman mapping images of the transferred graphene samples on SiO_2/Si . All the samples were implanted with $1 \times 10^{16}\text{ ions cm}^{-2}$, corresponding to nominal bilayer graphene. (A colour version of this figure can be viewed online.)

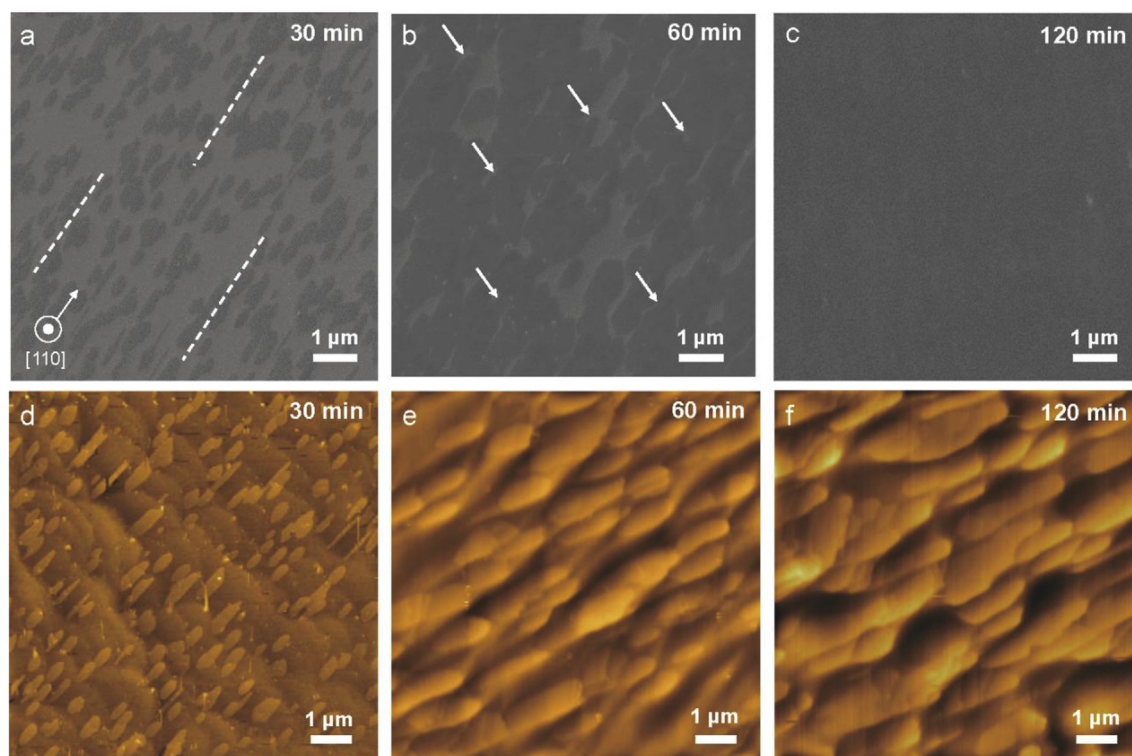


Fig. 3. (a–c) SEM and (d–f) AFM images of graphene islands at the different stages of growth with implanted fluence of 1×10^{16} ions cm^{-2} for 30 min, 60 min and 120 min at 900 °C. The white dashed lines in figure a show the same orientation of graphene islands. The white arrows in figure b point out the coalescence of the adjacent graphene islands. (A colour version of this figure can be viewed online.)

phenomenon observed by AFM measurement where all the graphene islands nucleation tends to occur near the Ge atomic step edges at the early stage of graphene growth, which is similar to that the seeds germinate from the step sites in Fig. 3d. This phenomenon was also observed on Ge substrate via CVD method [10], and the graphene with zigzag edge at atomic steps of Ge(110) is more stable according to the DFT calculation [20]. Fig. 3e–f show the graphene islands grow up to all over of the surface of Ge substrate. Moreover, it was also found that a pre-annealing under Ar/H₂ for 30 min followed by carbon implantation yielded more uniform graphene on the Ge surface because pre-annealing could significantly reduce steps density of Ge substrate leading to lower the activation barriers typically associated with diffusion process.

To further confirm that the feather-like islands shown in Fig. 3 are graphene islands, lateral force microscopy (LFM) was conducted to analyse the feather-like islands at high resolution. LFM is a simple and effective approach for identifying graphene at atomic scale by measuring the cantilever's lateral deflection in contact mode, which is independent of topography even on rough growth substrate [34]. Fig. 4 a–b show the topography images of the sample with implanted fluence of 1×10^{16} ions cm^{-2} with the growth time for 30 min. Fourier filtered lattice-resolution image of the graphene island on Ge substrate was obtained with LFM in air, as shown in Fig. 4c. A typical fast Fourier transform (FFT) pattern of the lattice-resolution image reveals a hexagonal atomic structure with the lattice constant $d = 0.244 \pm 0.02$ nm, close to the lattice constant of graphene $d = 0.246$ nm [35], as shown in Fig. 4d. Similar lattice-resolution images on other islands were also analysed indicating all the feather-like islands are graphene. Thus, high-crystallinity graphene islands covers the whole surface of the Ge substrate.

In order to further investigate the microscopic information such as thickness, stacking order and crystallinity of graphene, the transferred graphene on c-flat carbon grids was characterized by

transmission electron microscope (TEM). Fig. 5a shows a typical low-magnification TEM image of the graphene grown on Ge substrate with implanted fluence of 1×10^{16} ions cm^{-2} for 120 min at 900 °C. Fig. 5b shows the high-resolution TEM image obtained from the edges indicating a double layer of graphene. The selected area electron diffraction (SAED) pattern of graphene, shown in Fig. 5c, exhibits typical hexagonal diffraction, and the intensity of outer diffraction spots appears higher than that of inner diffraction spots, suggesting the formation of high-crystallinity Bernal stacking bilayer graphene. Similar graphene samples with different carbon ion implanted fluence were also characterized indicating that the layer of graphene can be tuned by the implanted fluence, as shown in Fig. S4.

Fig. 6a shows the Raman spectra of Ge substrates were implanted with 10 keV carbon ions at three different fluences: 5×10^{15} ions cm^{-2} , 1×10^{16} ions cm^{-2} and 2×10^{16} ions cm^{-2} . For the sample with the fluence of 5×10^{15} ions cm^{-2} , the symmetrical 2D band can be fitted by a single sharp Lorentzian peak with a full width at half maximum (FWHM) of ≈ 33 cm^{-1} , indicating the formation of the monolayer graphene [36]. For the samples with the fluence of 1×10^{16} ions cm^{-2} and 2×10^{16} ions cm^{-2} , the 2D bands can be fitted by four and six Lorentzian curves with FWHM of 44 cm^{-1} and 64 cm^{-1} , respectively, suggesting the formation of bilayer and multilayer graphene, as shown in Fig. S5. From the change of I_{2D}/I_G ratios and I_D/I_G ratios in Raman spectra, the number of graphene layers increases gradually and the defects of graphene decreases gradually with the increase of the carbon implanted fluence. Interestingly, the result of graphene growth on Ge substrate using ion implantation reveals that the growth process is not self-limiting, which is in contrast to previous observation for CVD method [37]. This suggests that the graphene growth mechanism by ion implantation is similar to that of the dissolution-segregation method [6,36,38], which has a dramatic difference with that by the

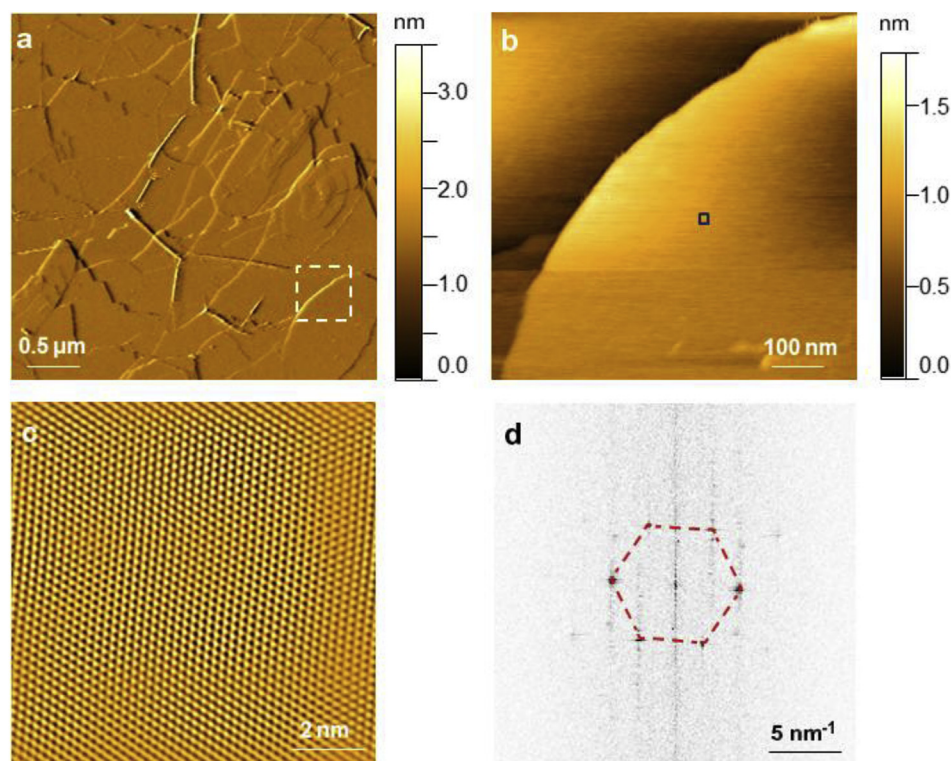


Fig. 4. (a) and (b) are topography images of graphene on Ge substrate. (b) was an enlarged image taken from the white dashed box marked in (a). (c) Fourier filtered lattice-resolution image (scan size 10 nm) of the graphene island marked by the blue box in (b), showing carbon lattices with 6-fold symmetry. (d) The FFT pattern was extracted from the pristine lattice-resolution image. The red dashed lines highlight a hexagonal pattern of graphene. (A colour version of this figure can be viewed online.)

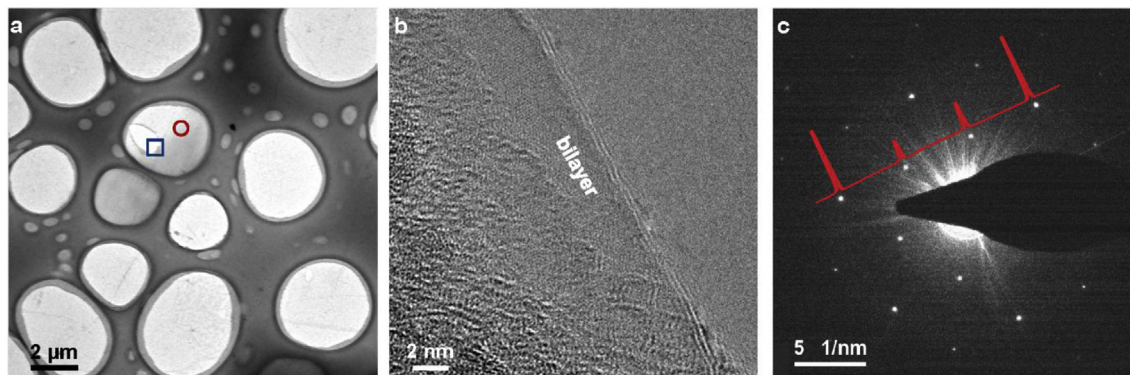


Fig. 5. (a) Plan-view TEM image of the graphene with implanted fluence of 1×10^{16} ions cm^{-2} for 120 min at 900 °C. (b) High-resolution TEM image of the crack edge of the bilayer graphene from the blue box. (c) The SAED pattern of bilayer graphene obtained from the red circle indicates the formation of high-crystallinity graphene. (A colour version of this figure can be viewed online.)

surface-mediated CVD method [5]. During annealing, the implanted carbon atoms diffuse to the surface of Ge substrate, and self-assemble into graphene under the catalysis of Ge. As the carbon atoms continue to segregate on the surface, the subsequent graphene layer nucleate underneath the first layer. The graphene formed earlier provides a flat plane for the subsequent graphene layer growth, which leads to fewer defects in multilayer graphene [39]. The movement of carbon atoms was driven by low carbon solid-solubility of Ge during the annealing process, which was supported by the change of carbon atomic concentration obtained from the XPS depth profiles (Fig. S6). Moreover, regardless of whether a slow-cooling process or fast-cooling process was adopted, the quality and layer of graphene were almost unchanged,

which means the growth process of graphene was at the steady annealing period and the graphene layer significantly depended on the implanted carbon fluence, as shown in Fig. S7.

Raman spectrum of the Ge(110) substrate shows the absence of Ge-C related peaks indicating there is no the bond between the carbon atoms and the underlying Ge surface [10], as shown in Fig. 6b. In order to clarify the carbon bonding nature, XPS measurements on Ge substrate were carried out. Fig. 6c shows the featured C 1s signals of graphene with an sp^2 carbon peak (284.8 eV), an sp^3 peak (285.6 eV), a C-O peak and a broad O-C=O peak, indicative of good quality of as-grown graphene on Ge substrate [40]. Moreover, the oxidation of Ge surfaces with and without graphene was investigated via the analysis of the evolution of Ge

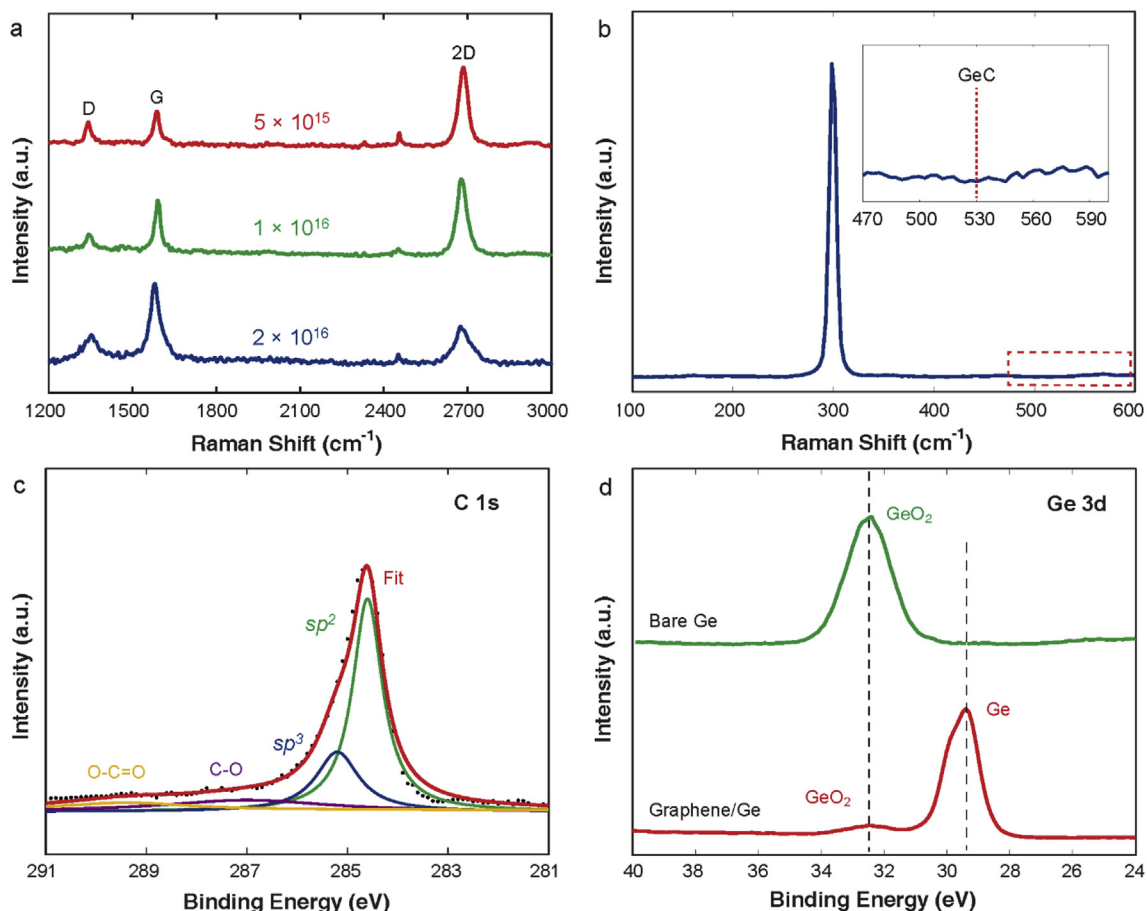


Fig. 6. (a) Raman spectra obtained from the graphene directly grown on Ge (110) with different carbon implanted fluences. (b) Raman spectrum of the Ge(110) substrate shows the absence of Ge-C related peaks. (c) XPS spectrum of C 1s showing the featured signals of graphene with an sp^2 carbon peak (284.8 eV), an sp^3 peak (285.6 eV), a C-O peak and a broad O-C=O peak, respectively. (d) XPS spectra of Ge 3d compare the bare Ge sample and graphene/Ge sample after exposure to air at room temperature for 15 days. (A colour version of this figure can be viewed online.)

photoelectron spectra over time. Fig. 6d compares the Ge 3d spectra of bare Ge sample removing oxidation layer and graphene/Ge sample after exposure to air at room temperature for 15 days. The GeO_2 layer will form on the surface of the bare Ge caused by the oxidizing species (OH , O_2) in the air for 15 days [41]. When there exists graphene on Ge surface, oxide-related peak is dramatically reduced in contrast to that on bare Ge indicating that graphene appears to act as a screen to effectively hinder Ge surface from oxidation, which possesses potential applications in preventing oxidation in semiconductor industry [41].

4. Conclusion

In summary, a facile and transfer-free approach was demonstrated to synthesize graphene on Ge(110) substrate. By taking advantage of precisely controlled carbon ion implantation, the carbon atoms can segregate on the catalyst surface from the near-surface substrate and form graphene with the predetermined layer number during annealing. Our results reveal that the graphene nucleation tends to occur near the Ge atomic step edges upon carbon segregation process. In addition, graphene growth via ion implantation is not self-limiting different from the previous observation in CVD experiment. Thus, the ion implantation technique can be exploited not only to produce layer-tunable graphene by controlling carbon implanted fluence but also to open up a new pathway to understand the growth process of graphene.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbon.2019.07.029>.

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